

## Steps

## Reason

$$1) \forall x (O(x) \Rightarrow R(x) \vee S(x))$$

Premise

$$2) Su(Tuesday)$$

Premise

$$3) Su(Tuesday) \Rightarrow \neg R(Tuesday)$$

Premise

$$4) \neg R(Tuesday) \wedge \neg S(Tuesday)$$

Modus ponens on 2, 3

$$5) \neg (R(Tuesday) \vee S(Tuesday))$$

2nd De Morgan's Law on 4

$$6) O(Tuesday) \Rightarrow R(Tuesday) \vee S(Tuesday)$$

Universal instantiation on 1

$$7) \neg O(Tuesday)$$

Modus tollens on 5, 6

$$8) O(Tuesday) \vee O(Thursday)$$

Premise

$$9) O(Thursday)$$

Disjunctive syllogism on 8, 7

$$10) O(Thursday) \Rightarrow R(Thursday)$$

Universal instantiation on 1

$$11) R(Thursday) \vee S(Thursday)$$

Modus ponens on 9, 10

$$12) \neg S(Thursday)$$

Premise

$$13) R(Thursday)$$

Disjunctive syllogism on 11, 12

$$14) R(Thursday) \vee S(Thursday)$$

$$15) R(Thursday)$$

$$16) R(Thursday)$$

$$17) R(Thursday)$$

$$18) R(Thursday)$$

$$19) R(Thursday)$$

$$20) R(Thursday)$$

$$21) R(Thursday)$$

$$22) R(Thursday)$$

$$23) R(Thursday)$$

$$24) R(Thursday)$$

$$25) R(Thursday)$$

$$26) R(Thursday)$$

$$27) R(Thursday)$$

$$28) R(Thursday)$$

$$29) R(Thursday)$$

$$30) R(Thursday)$$

$$31) R(Thursday)$$

Valid Conclusions: 1) I did not take Tuesday off 2) I took Thursday off

3) It rained on Thursday.

b) "If I eat spicy foods, then I have strange dreams." "I have strange dreams if there is thunder while I sleep." "I did not have strange dreams."

p: I eat spicy food. q: I have strange dreams r: There is thunder while I sleep.

[Soln:] p: I eat spicy food. q: I have strange dreams r: There is thunder while I sleep.

[Premises:] 1)  $p \rightarrow q$  2)  $r \rightarrow q$  3)  $\neg q$